Diagnose Before Solving: Key Decisions

David Bacon · November 2026

The full decision log runs to more than 200 dated entries. Each records what was decided, the alternatives, and the reasoning, and entries are never rewritten: a reversal is a new entry pointing back at the old one. This page picks out the twenty-two decisions that best show how the project was thought through, grouped by the question they answer. Each is listed by its D-number, so it can be found in the full log.

Why rebuild at all

#	Decision	Why it matters
D-001	Replace the third-party-hosted simulation with a self-hosted build	The original failed in front of a prospective employer when its credits ran out. Every reliability decision below traces back to this one
D-004	Hard spending caps, with a recorded demo instead of an error page	An "unavailable" message reproduces the original failure in a politer font. A visitor must always see a complete artifact
D-005	The design documents are deliverables, not overhead	For an instructional design portfolio, reasoning is stronger evidence than output

How Marcus works

#	Decision	Why it matters
D-029	Marcus's knowledge is tiered, and the root cause is confirm-only	Unconstrained, an AI playing a frustrated user explains its own problem by the second message, and there is nothing left to diagnose. This is the load-bearing decision of the whole design
D-038	Marcus arrives with a half-right suspicion he can't explain	A truer and harder task: the learner must take his theory seriously without accepting it. He also commits the very mistake the course is named for
D-040	Marcus opens up only when he feels heard, not just when asked well	Makes the rubric mechanically true: a learner who diagnoses before acknowledging doesn't just lose points, they don't get the information
D-041	"I understand, but..." makes Marcus angrier	An acknowledgment used as a preamble to a pivot reads as a technique. Acknowledgment followed by a real question is rewarded, so the rule is about the pattern, not the word "but"
D-043	Recovery from a deep hole is asymmetric	Arithmetic on paper showed a learner who blundered by turn 9 could not win with perfect play. Fixed before any code existed, and later confirmed by a transcript that recovered in eight turns

What the learner doesn't see

#	Decision	Why it matters
D-055	No rubric on screen before or during the call	A learner who can see the criteria performs a checklist. The rubric appears in the results, where it teaches
D-081	No turn counter; Marcus signals the call is dragging himself	Real calls have no timer, and reading someone's patience is part of the skill. The warning moves inside the fiction rather than disappearing

How scoring stays fair and consistent

#	Decision	Why it matters
D-008, D-049	Separate character and assessor models, and the assessor never sees the character's internal state	The character must not grade the learner, and every judgment must be checkable against the transcript alone
D-050	The model judges; the code counts	Given the same judgments, the score is identical every time. Removes an entire class of error from the project's highest-rated risk
D-	A consistency failure traced to the rubric,	One criterion contained two incompatible tests. The model resolved the

110	not the model	contradiction two defensible ways, six minutes apart
D-129	Full credit for follow-up requires volunteering it, judged by turn order	Decided by counting rather than by tone, because "did they volunteer it" is exactly the kind of standard that scores differently each run
D-132	Consistency evidence measures the instrument, not the standard	Ten identical results showed one prompt resolving an ambiguity consistently, not that the ambiguity was gone. It tipped as soon as a different criterion was edited
D-140	The fifth rubric ambiguity, and the pattern behind all five	Each time one boundary was made explicit, the next most ambiguous one began to move. A rubric written for human readers does not survive being applied literally and repeatedly

How the testing was run

#	Decision	Why it matters
D-071	The test plan was written before the build	Written after, a test plan describes what the build does. Written before, it holds the uncomfortable tests: does Marcus leak the answer, does the score move
D-094	Verify as an anonymous stranger, not from the owner's browser	"It loads fine when I click it" is exactly the condition under which the original failed
D-126, D-141	Feedback now survives a reload; the defect was found by a thumb	A swipe-back on a phone destroyed the feedback permanently. No automated test could have found it, which is the case for the human pilot in one sentence
D-195, D-196	Two serious defects found between two correct tests	A refused session inside an embedded frame looped about 80 requests a second and showed the visitor nothing. Each test was right on its own; the combination had never been run

Delivery

#	Decision	Why it matters
D-173, D-214	The lead-in training is built outside an authoring tool	Storyline work is already shown elsewhere in the portfolio; a complete experience in plain HTML, CSS and JavaScript shows something different
D-188	Embedding is kept, and verified against a test host	The simulation stays embeddable and LMS-mappable by design, with the claim stated no more strongly than the evidence supports

---

Where each entry lives. Part 1 covers D-001 to D-145, Part 2 D-146 to D-172, Part 3 D-173 to D-188, Part 4 D-189 to D-193, Part 5 D-194 to D-203, Part 6 D-204 to D-209, Part 7 D-210 to D-212, and Part 8 D-213 onward.